

#7 Vehicle Time Management

Objective

To design a system for managing and displaying vehicle date and time

Description

The system uses a Real-Time Clock (RTC) module integrated with the STM32F405RGT6 microcontroller to keep track of the current date and time. Input switches are used to set or update the date and time, and the values are displayed on a 16x2 LCD.

Hardware Required

1. CORTEX M4-STM32F405 Board,
2. DS1302 RTC Real Time Clock Module with Battery
3. LCD
4. Onboard Push Button Switches

Software Required

1. STM32 CUBE IDE

Expected Result & Test Cases

Test Case 1: Setting RTC Current Date and Time in 12-hour Format with AM/PM Display

Objective: Test the functionality of setting the RTC to a current date and time in a 12-hour format (with AM/PM) using input switches.

Steps:

1. Power ON the system.
2. Press the **first switch** to start setting the date, time and day. The cursor will be at the first column, first row. For example:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
<u>D</u>	:	3	0	-	1	2	-	2	0	2	5	,	W	E	D
T	:	1	2	F	M	T		1	2	:	0	0		P	M

3. Press the **first switch** repeatedly to increment the current digit/letter
 - For example – Column 1, Row 1 can iterate through D,T
 - For example – Column 3, Row 1 can iterate through 0 – 3
 - For example – Column 4, Row 1 can iterate through 0 – 9
 - For example – Column 14, Row 1 can iterate through M,T,W,F,S
 - For example – Column 15, Row 1 can iterate through O,U,E,H,R,A
4. Press the **second switch** to confirm the current digit/letter and move to the next column.
5. If the **second switch** is pressed when the cursor is in the last column, the cursor should again cycle to the first column.
6. Use the **third switch** to move the cursor to the next row of the same column.
7. Repeat the above process for the month, year, hour, and minute fields.
 - Adjust 31-12-2024 11:59 PM.
8. Once all fields are adjusted, press the **fourth switch** to confirm the new date and time.
9. The LCD should be updated to display the newly set date and time in the 12-hour format with AM/PM. As shown below without any cursor

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
<u>D</u>	:	3	0	-	1	2	-	2	0	2	5	,	W	E	D
T	:	1	2	F	M	T		1	2	:	0	0		P	M

Expected Result:

- The RTC date and time are correctly set by the user in the 12-hour format with AM/PM, and the LCD reflects the updated time immediately.
- If the user sets the time close to midnight (e.g., 11:59 PM on 12-31-2024), the year, date, and time should roll over correctly to 01-01-2025 12:00 AM.

Test Case 2: Power Cycle and Verify RTC Retention in 12-hour Format with AM/PM Display

Objective: Verify that the RTC retains the set date and time in 12-hour format with AM/PM after a power cycle.

Steps:

1. Power ON the system and verify the displayed date and time on the LCD (e.g., 12-31-2024 11:59 PM).
2. Power OFF the system.
3. Wait for a brief period (e.g., 10-15 seconds).
4. Power ON the system again.
5. Observe the LCD display.

Expected Result:

- The LCD should display the same date and time as was set before the power cycle (e.g., 12-31-2024 11:59 PM).
- The RTC successfully retains the time due to the backup battery.
- The year, date, and time should correctly roll over if set to the last day and hour (e.g., 12-31-2024 11:59 PM should transition to 01-01-2025 12:00 AM).

Test Case 3: Set Time to Actual Date and Time in 24-hour Format

Objective: Set the RTC date and time to the actual current time in 24-hour format using input switches.

Steps:

1. Power ON the system.
2. The system will display the date and time that was previously set (e.g., 12-31-2024 11:59 PM).
3. Press the **first switch** to start setting the date, time and day. The cursor will be at the first column, first row, as shown below.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
<u>D</u>	:	3	0	-	1	2	-	2	0	2	5	,	W	E	D
T	:	2	4	F	M	T		2	3	:	1	5			

4. Press the **second switch** to confirm the current digit/letter and move to the next column.
5. The user will be allowed to set the time in the 24-hour format only when Column 3 & 4 of 2nd Row is set to 24. As shown below

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
D	:	3	0	-	1	2	-	2	0	2	5	,	W	E	D
T	:	2	4	F	M	T		2	3	:	1	5			

6. Once the user selects the 24-hour format, the AM/PM fields should become empty automatically and the range for columns 9 and 10 of 2nd row also needs to be updated accordingly.
7. If the **second switch** is pressed when the cursor is in the last column, the cursor should again cycle to the first column.
8. Use the **third switch** to move the cursor to the next row of the same column.
9. Adjust the day, month, year, hour, and minute using the switches to match the current actual time in 24-hour format.
10. Once all fields are adjusted, press the **fourth switch** to confirm the new date and time.
11. The LCD should be updated immediately to display the new date and time (e.g., 01-01-2025 14:30).

Expected Result:

- The RTC is successfully set to the current date and time in the 24-hour format, and the LCD reflects the updated time immediately.

Test Case 4: Ensure Accurate Time and Date After Power Cycle in 24-hour Format

Objective: Ensure that the RTC keeps accurate time and the display on the LCD is correct in 24-hour format even after a power cycle.

Steps:

1. Power ON the system and verify that the correct date and time is displayed on the LCD in 24-hour format (e.g., 01-01-2025 14:30).

2. Check the actual current date and time using a reliable reference (e.g., smartphone or computer).
3. Power OFF the system.
4. Wait for about 15 minutes.
5. Power ON the system again.
6. Observe the LCD display and verify that the displayed time matches the actual time.

Expected Result:

- The date and time displayed on the LCD should match the actual current time in 24-hour format.
- The RTC continues to keep accurate time, and the system displays the correct date and time after power cycles.
- The year, date, and time should roll over correctly at the end of the year or the last day of the month (e.g., 12-31-2024 rolls over to 01-01-2025 correctly, even after power cycles).

1st Switch: For Editing the Date and Time Fields

2nd Switch: For Confirming the current digit/letter and moving to the next column. Will Roll Back.

3rd Switch: For Moving the Cursor to the next Row of the same Column. Will Roll Back.

4th Switch: For Confirming the Set Date and Time for proceeding with RTC Write.

Messaging

- Every 1 Min Send the Following Time Stamp Data to UART as per the Format Set in the LCD.
- D: 31-12-2025, WED
- T: 12:15 PM
- Or
- D: 31-12-2025, WED
- T: 22:15

Project Deliverables

1. Project Report containing
 - i. Flowchart

- ii. H/w Circuit - Photos
- iii. Code
- iv. Output Screenshots/Photos
- v. Live Expressions - Screenshots

2. Project Report to be Uploaded into LMS

Project Assessment

Project Activities	Weightage (%)
H/w Setup	25%
Coding	25%
Testing	25%
Documentation	25%